CB23F35 · LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

EviTrace

Digital Evidence Chain-of-Custody and Investigation Tracking System



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THE PROBLEM

Evidence tracking is scattered today

Fragmented records

Case, evidence, custody and examination data sit in different files and cannot be traced from one to another.

Weak custody history

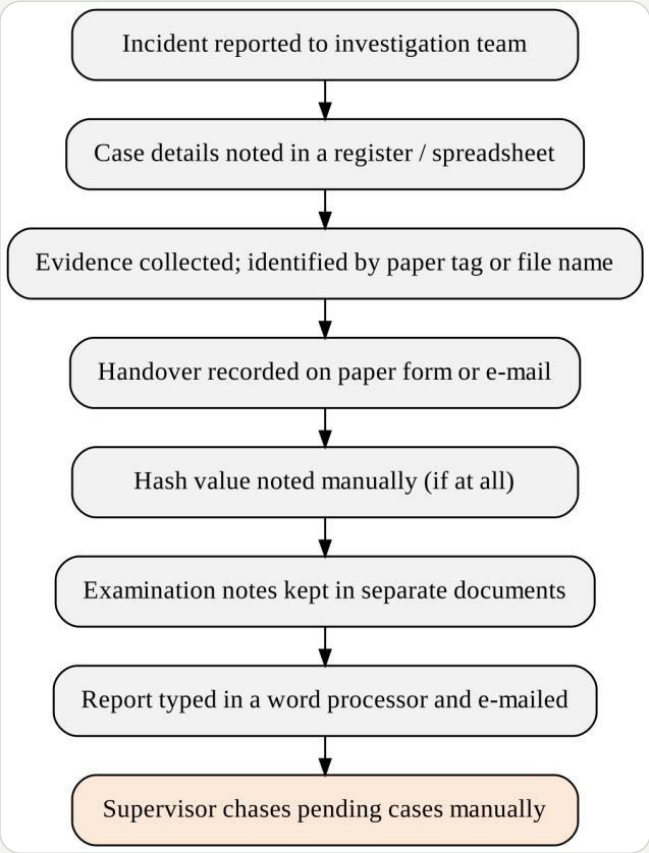
Hand-overs are recorded late, incompletely or not at all, so “who held it, and when?” has no quick answer.

Unrecorded integrity checks

Hash values are noted by hand, if at all, and cannot show that evidence is unchanged.

Manual chasing

Supervisors chase old cases and draft reports by hand, with no automatic reminders or approval.



THE SOLUTION

Records, reports and one dashboard

7 Linked forms

WHAT IT CONTAINS

Cases, Evidence, Chain of Custody, Integrity Verification, Forensic Examination, Reports and Audit Trail.

WHAT IT RESOLVES

Scattered records and weak custody history. Every item has one traceable record, and hash checks and examinations are stored against the evidence.

14 Reports

WHAT IT CONTAINS

7 lists, 2 kanban boards, a timeline and 4 pivots, with actions to run OCR and AI from the list.

WHAT IT RESOLVES

Manual chasing. Status, priority and event trends are visible without opening each record.

1 Command Center

WHAT IT CONTAINS

A dashboard page with 8 visuals: five chart types, a gauge, a KPI panel and a case board, plus open-by-case-ID.

WHAT IT RESOLVES

No single view of the workload. Officers see totals and trends in one place.

THE SOLUTION

Automation, access and smart assistance

7 Deluge functions and workflows

WHAT IT CONTAINS

5 active form workflows, a daily schedule, an auditor approval and a 3-step Blueprint, powered by 7 Deluge functions.

WHAT IT RESOLVES

Unrecorded events and late follow-up. Audit entries write themselves, old open cases trigger an e-mail and closure is validated.

4+1 Role profiles

WHAT IT CONTAINS

Investigation Officer, Evidence Custodian, Forensic Examiner and Auditor, plus a portal viewer profile.

WHAT IT RESOLVES

Uncontrolled access. Each role sees and edits only what its job needs, and reports need auditor approval.

3 Integrations

WHAT IT CONTAINS

Zoho CRM for organisation and contact, AI OCR for text from evidence files, Zia GenAI for report drafts and analysis.

WHAT IT RESOLVES

Duplicate data entry and slow report writing. Drafts are made for review and never final.

OBJECTIVES

What the project set out to do

01

Design a normalised data model and forms for all seven record types

02

Automate audit logging, reminders, case lifecycle and report approval

03

Give each role its own access, reports and a Command Center dashboard

04

Link each case to an existing organisation and contact through Zoho CRM

05

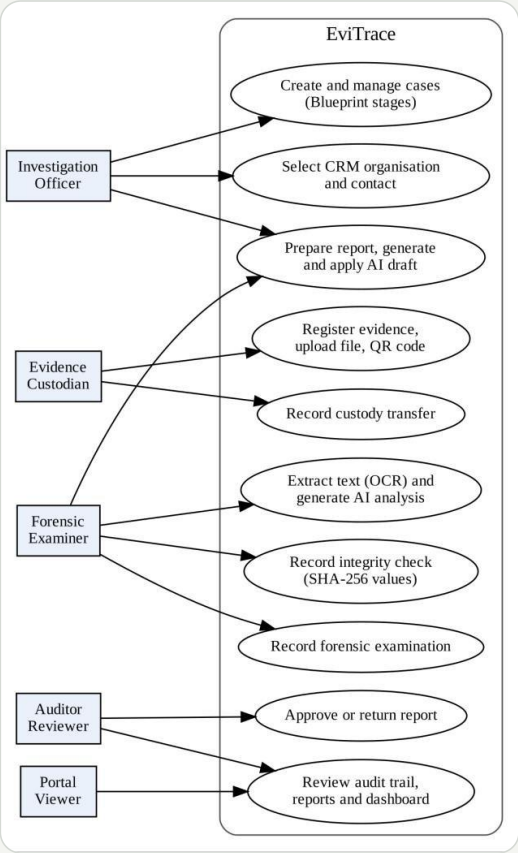
Use Zia and OCR to draft reports and analysis, with the investigator deciding

06

Deploy securely to production with roles, audit trail and approval

USERS AND ROLES

Five kinds of users, one shared record



Investigation Officer

Opens and manages cases, picks the CRM organisation and contact, prepares reports.

Evidence Custodian

Registers evidence, uploads files and records every custody transfer.

Forensic Examiner

Records integrity checks and examinations, runs OCR and AI analysis.

Auditor Reviewer

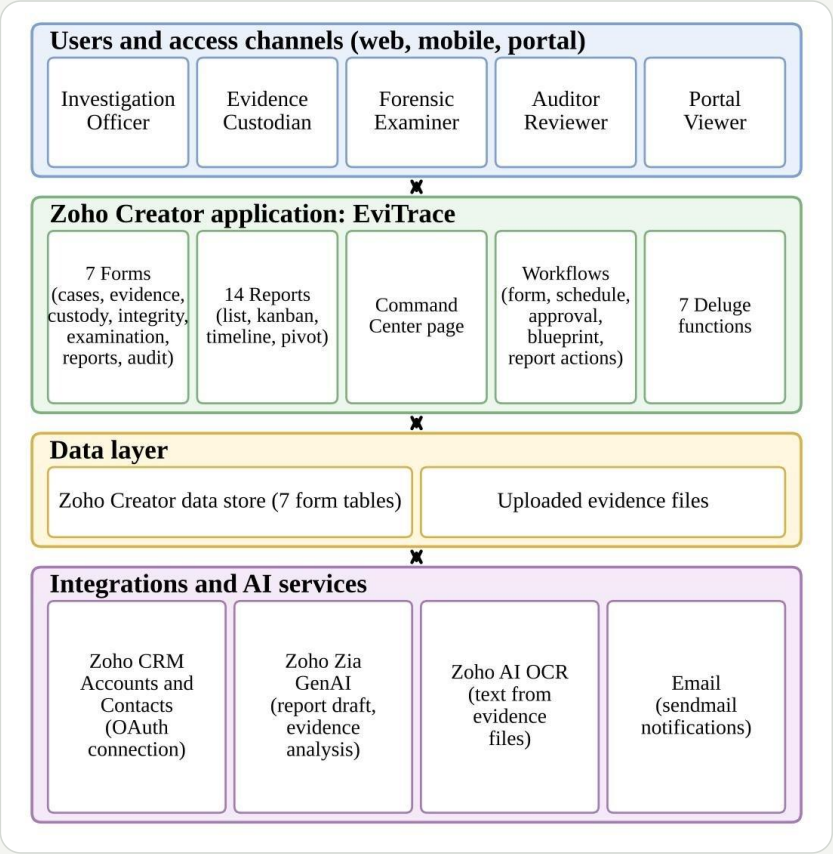
Approves or returns reports and reviews the audit trail.

Portal Viewer

Restricted view of authorised information through the customer portal.

SYSTEM ARCHITECTURE

Four layers on Zoho Creator



Users and channels

Web, mobile app and customer portal for the five roles.

EviTrace application

7 forms, 14 reports, Command Center page, workflows and 7 Deluge functions.

Data layer

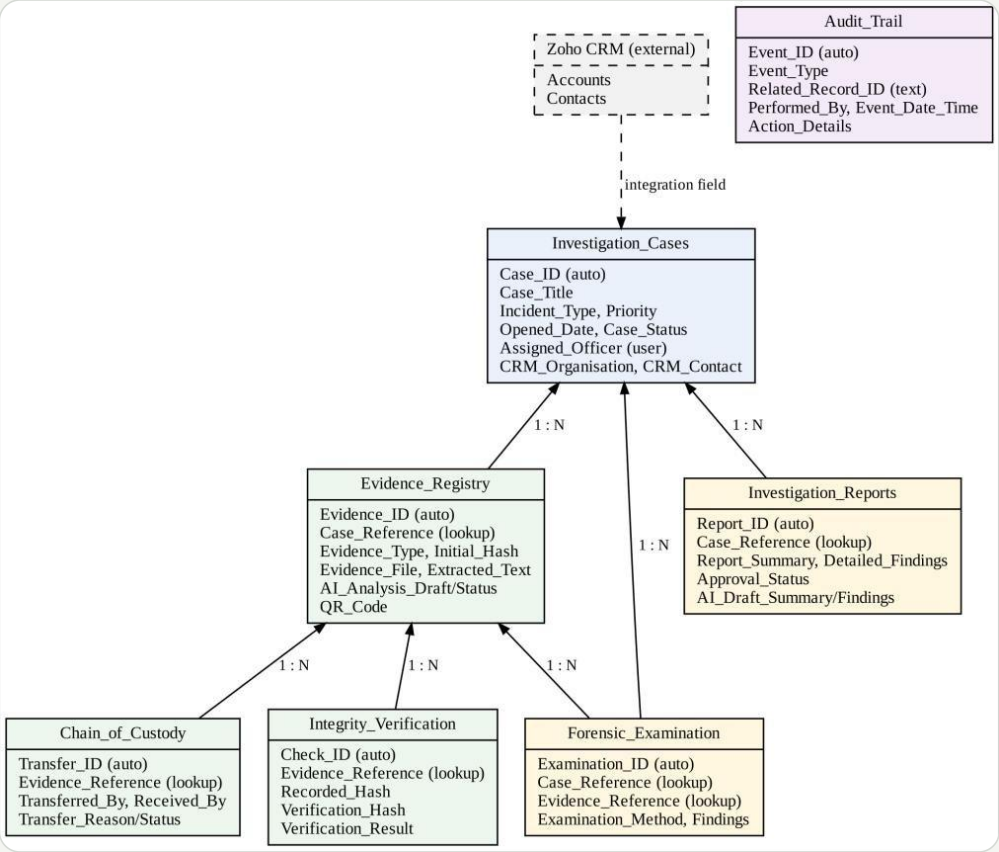
Creator data store for the form tables, plus uploaded evidence files.

Integrations and AI

Zoho CRM over OAuth, Zia GenAI for drafts, AI OCR for text, e-mail notifications.

DATA MODEL

Seven forms linked by lookups



Cases own evidence

Investigation_Cases links 1:N to Evidence_Registry and Investigation_Reports.

Evidence has a history

Custody transfers, integrity checks and examinations all point back to the evidence item.

CRM stays external

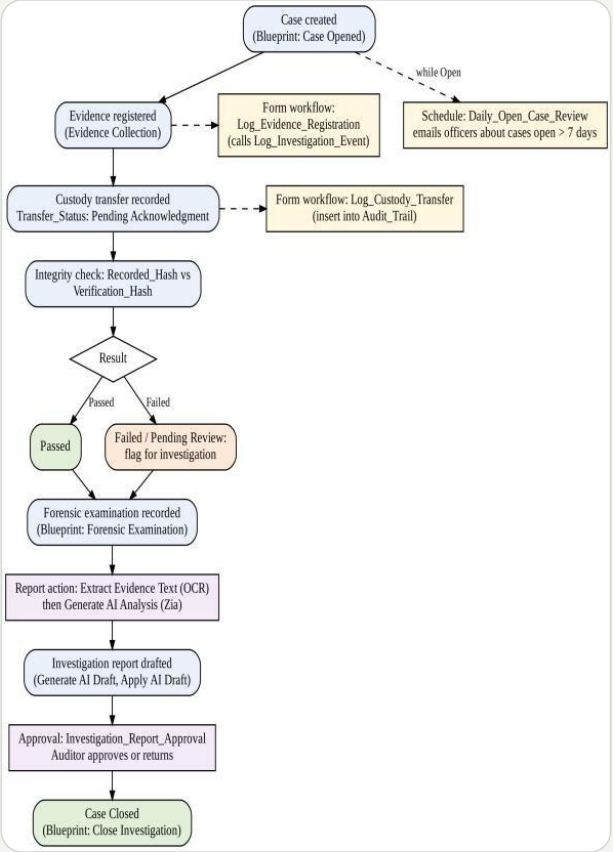
Organisation and contact are read from Zoho CRM, not copied into a new table.

Audit_Trail logs events

Written by workflows and the Log_Investigation_Event function.

PROCESS DESIGN

The evidence and case lifecycle



1 · Case opened

Blueprint starts at Open; a daily schedule e-mails the officer if a case stays open for more than 7 days.

2 · Evidence and custody

Each registration and transfer writes an audit entry automatically.

3 · Integrity and examination

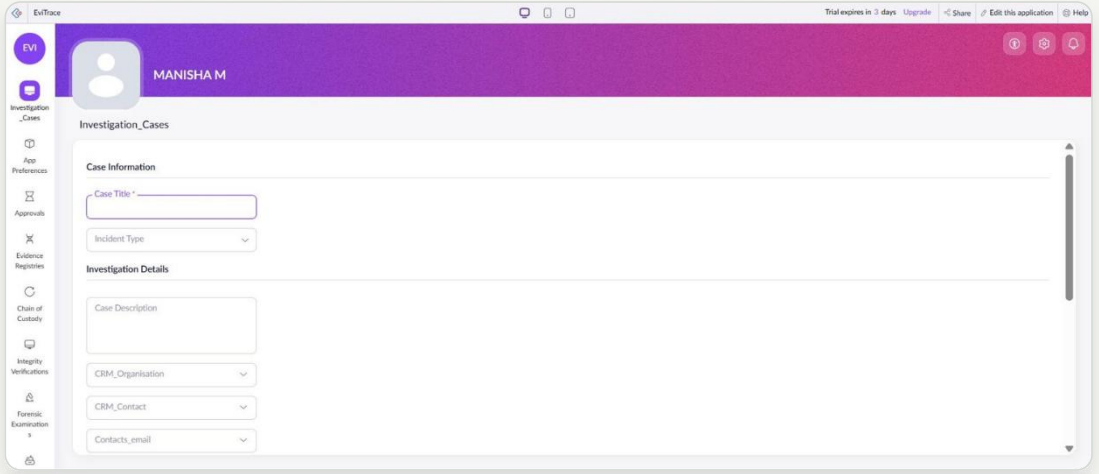
Recorded hash is compared with the verification hash; a failed check is flagged for investigation.

4 · Report, approval, closure

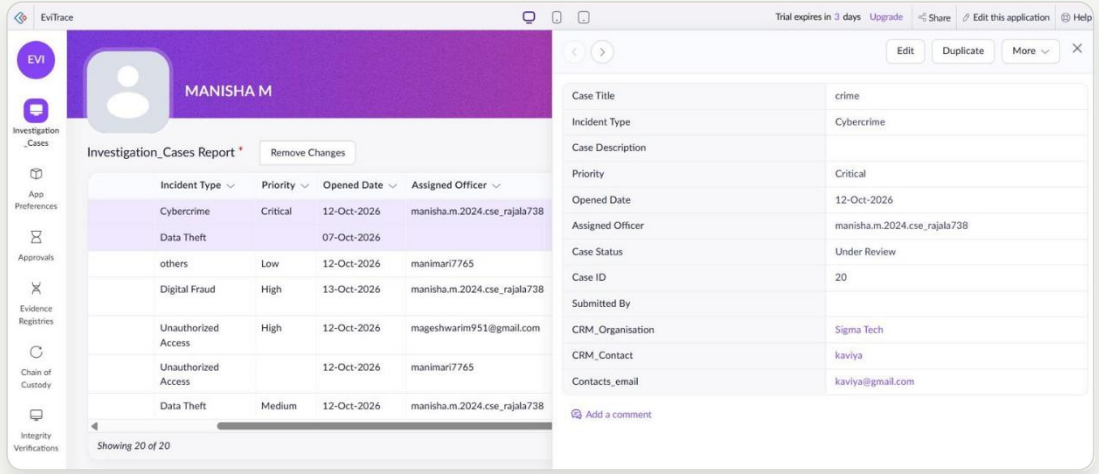
AI draft, then auditor approval, then the Close Investigation transition.

IMPLEMENTATION · FORMS

Forms capture every step



Investigation_Cases form, with CRM organisation and contact lookups



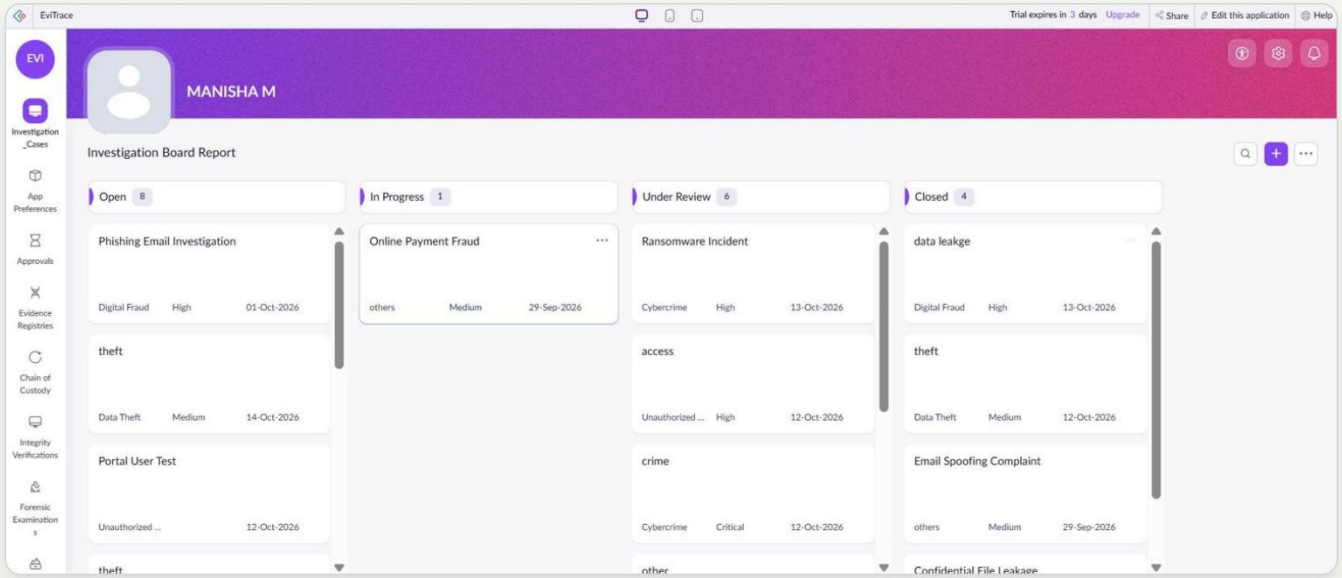
Saved case showing Sigma Tech and Kaviya M from Zoho CRM

Auto Number IDs **Lookups between forms** **File upload and QR code on evidence**

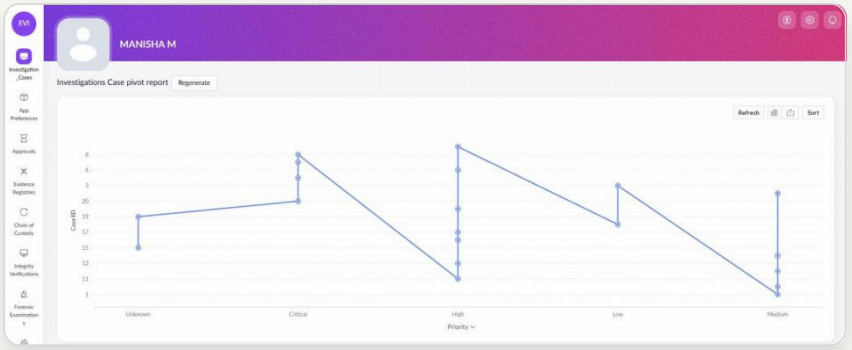
IMPLEMENTATION · REPORTS

14 reports in five types

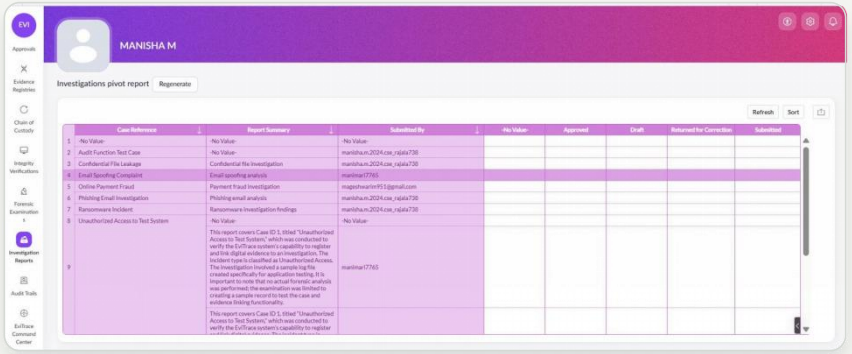
7 List | 2 Kanban | 1 Timeline | 2 Pivot chart | 2 Pivot table



Investigation Board: cases grouped by status



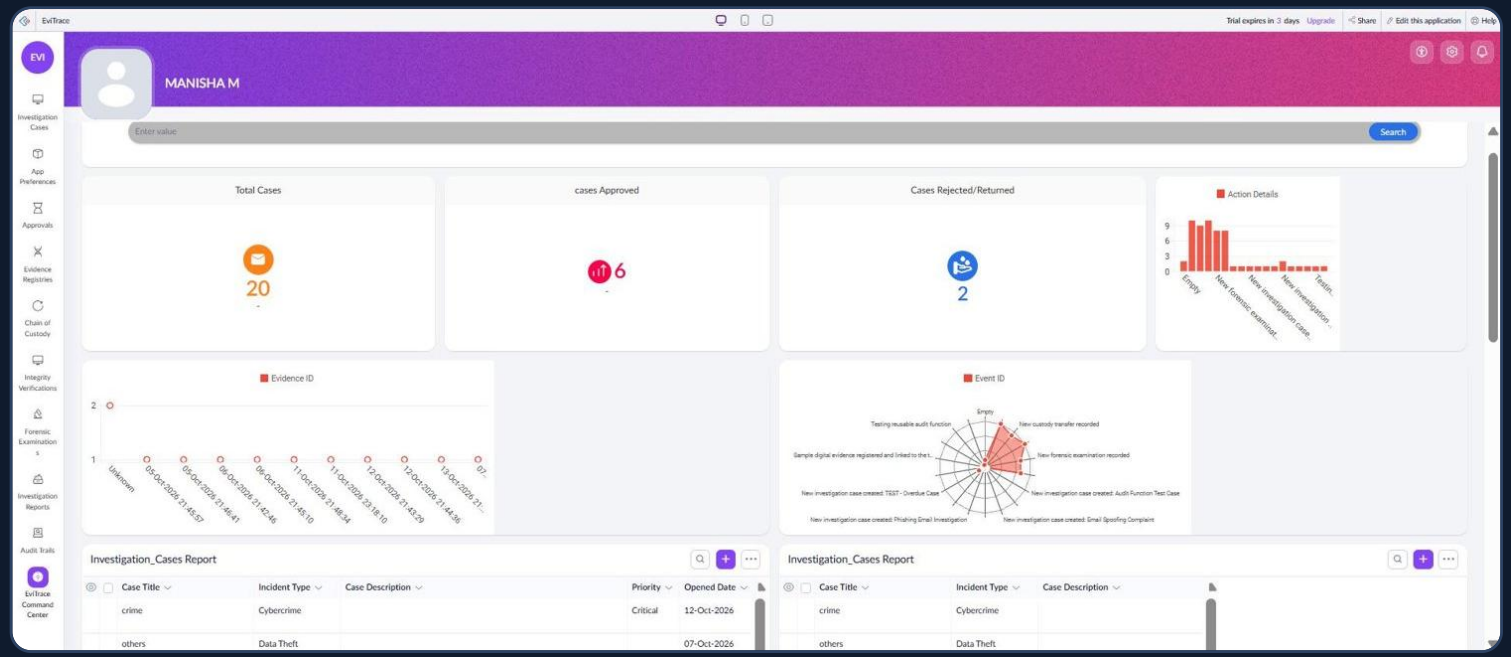
Audit timeline by event type



Pivot summary of investigations

IMPLEMENTATION · DASHBOARD

The EviTrace Command Center



Eight visual elements

Column, scatter, radar, line and donut charts, a gauge, a KPI panel and a board.

Open any case by ID

One field jumps to a case record.

IMPLEMENTATION · AUTOMATION

Workflows do the routine work

Form workflows

Write an audit entry on each registration, transfer, check and examination.

Schedule

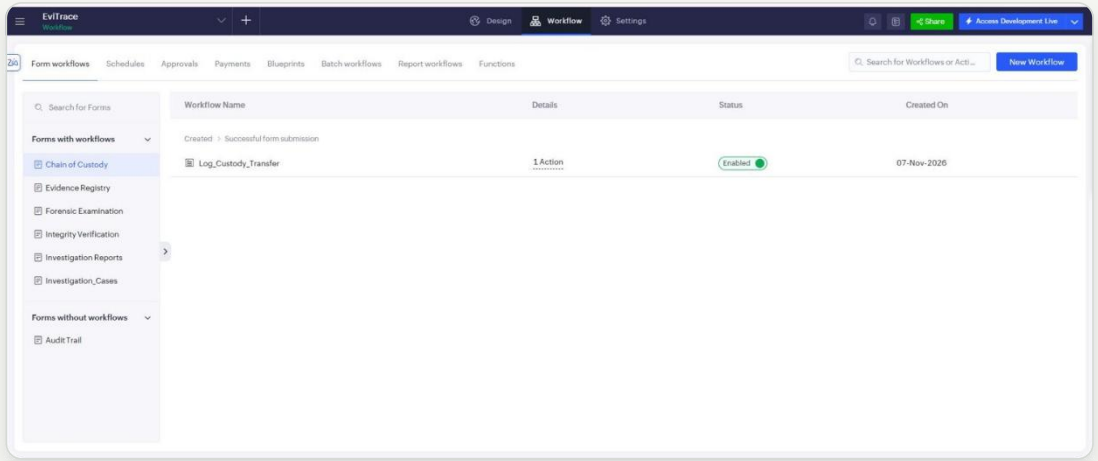
Daily e-mail to the officer for cases open over 7 days.

Approval

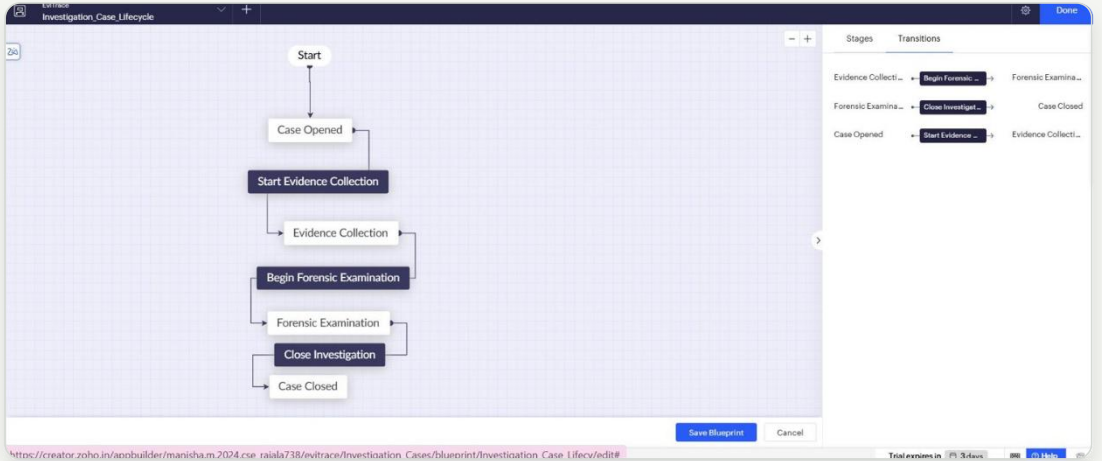
Submitted reports go to the Auditor to approve or return.

Blueprint

Three transitions control the case lifecycle.



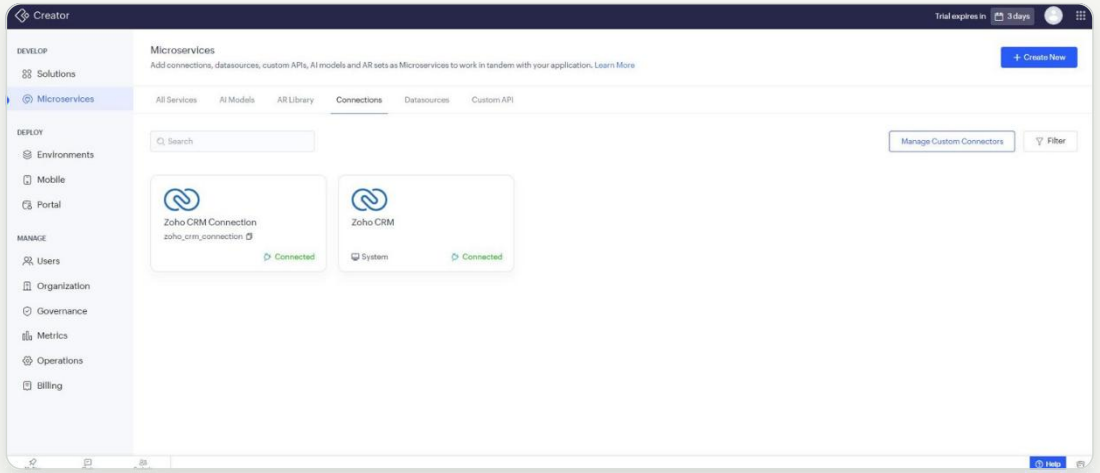
Workflows in the builder



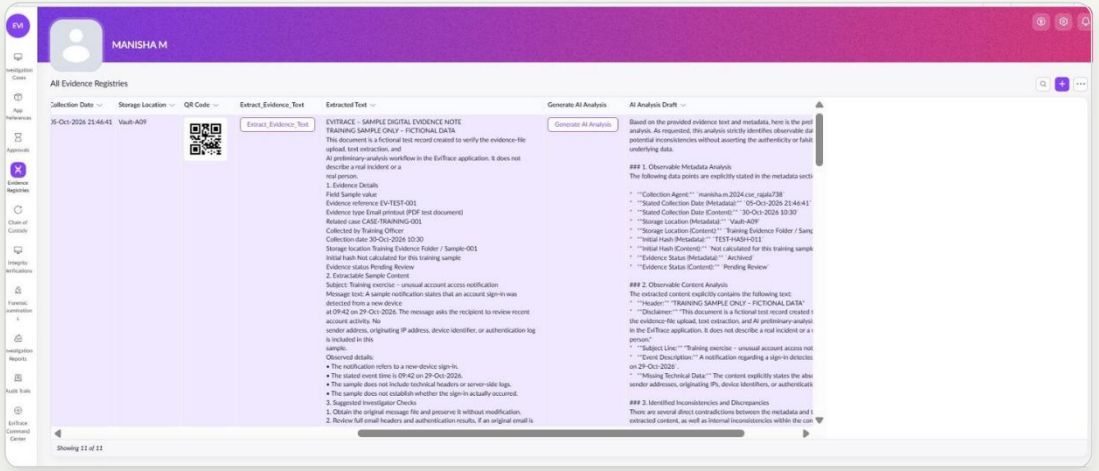
Investigation_Case_Lifecycle Blueprint

IMPLEMENTATION · INTEGRATIONS AND AI

CRM, OCR and AI drafts



zoho_crm_connection with Accounts and Contacts



OCR text and AI analysis draft on an evidence record

Zoho CRM

OAuth connection links each case to an organisation and contact.

Zoho AI OCR

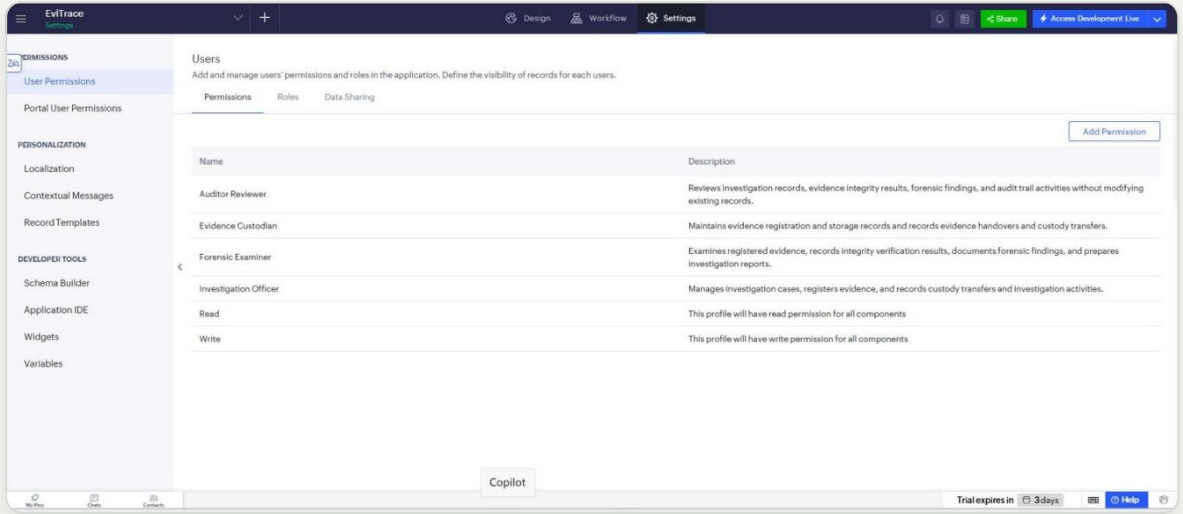
Extracts text from uploaded evidence files.

Zia GenAI

Drafts reports and analysis. Output stays a draft until a human applies it.

IMPLEMENTATION · SECURITY

Access control and safeguards



Four internal profiles plus EviTrace_Portal_Viewer

- **Role profiles: each role sees and edits only its own records**
- **Personal-data flag on user fields**
- **OAuth connection with limited CRM scopes**
- **Approval before a report counts as approved**
- **Audit trail of key events**
- **AI guardrails: drafts only, human review**

TESTING AND DEPLOYMENT

Tested, then published to production

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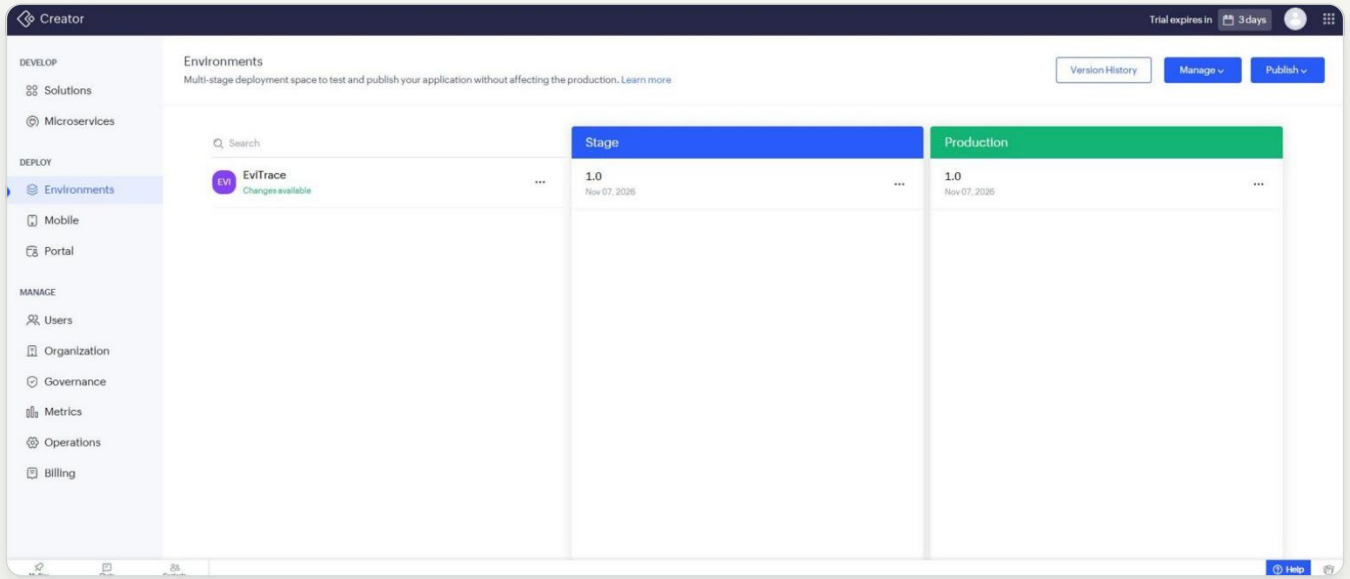
test cases passed, each mapped to a functional requirement

2

defects found and closed, including a portal-submission workflow issue

v1.1 live

Development, Stage, Production; deployed 07 November 2026

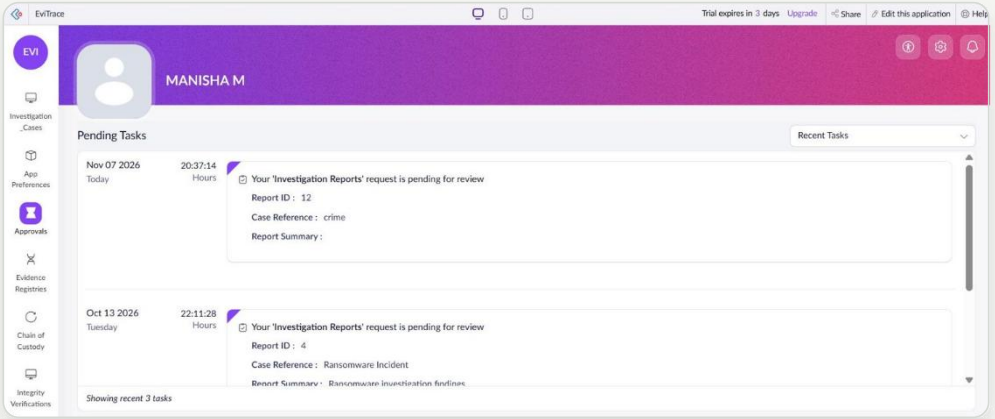


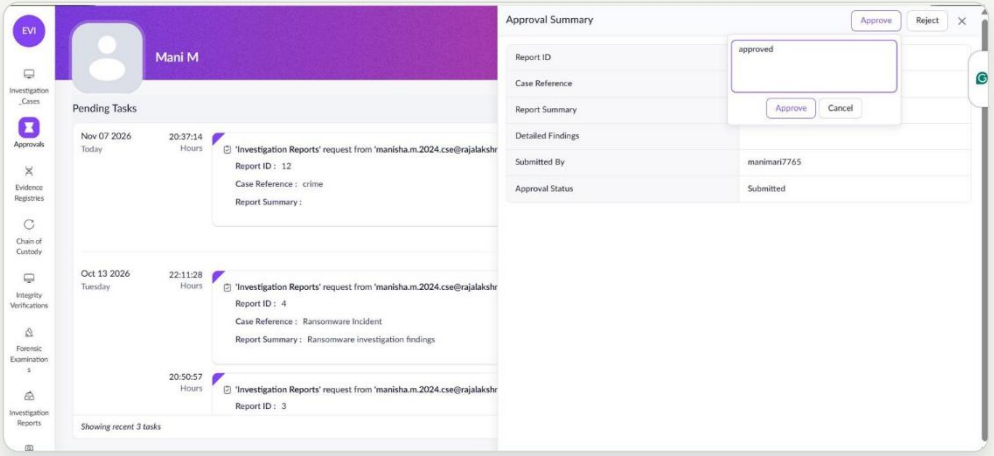
Environments: Development to Stage to Production

RESULTS · DEMO SCENARIO

One case, end to end

1. Officer creates a data-theft case for Sigma Tech, contact Kaviya M, from CRM
2. Custodian registers a laptop with its initial hash and storage location
3. Custodian records the transfer to the Forensic Examiner
4. Examiner records the integrity check and examination, runs OCR and AI analysis
5. Examiner generates and applies the AI report draft and submits it
6. Auditor approves the report; dashboard and audit timeline update





CONCLUSION

What was achieved and what comes next

Achieved

- One traceable record per case and evidence item
- Automatic audit entries and daily reminders
- Role-based access with auditor approval
- CRM-linked cases and AI-assisted drafting that cuts manual writing
- Built by one developer in one semester on a low-code platform

Future enhancements

- Compute SHA-256 of uploaded files automatically, for example with a Catalyst function
- Add record-level criteria for portal users
- Re-enable case-creation logging after fixing the portal issue

EVITRACE

Thank you

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